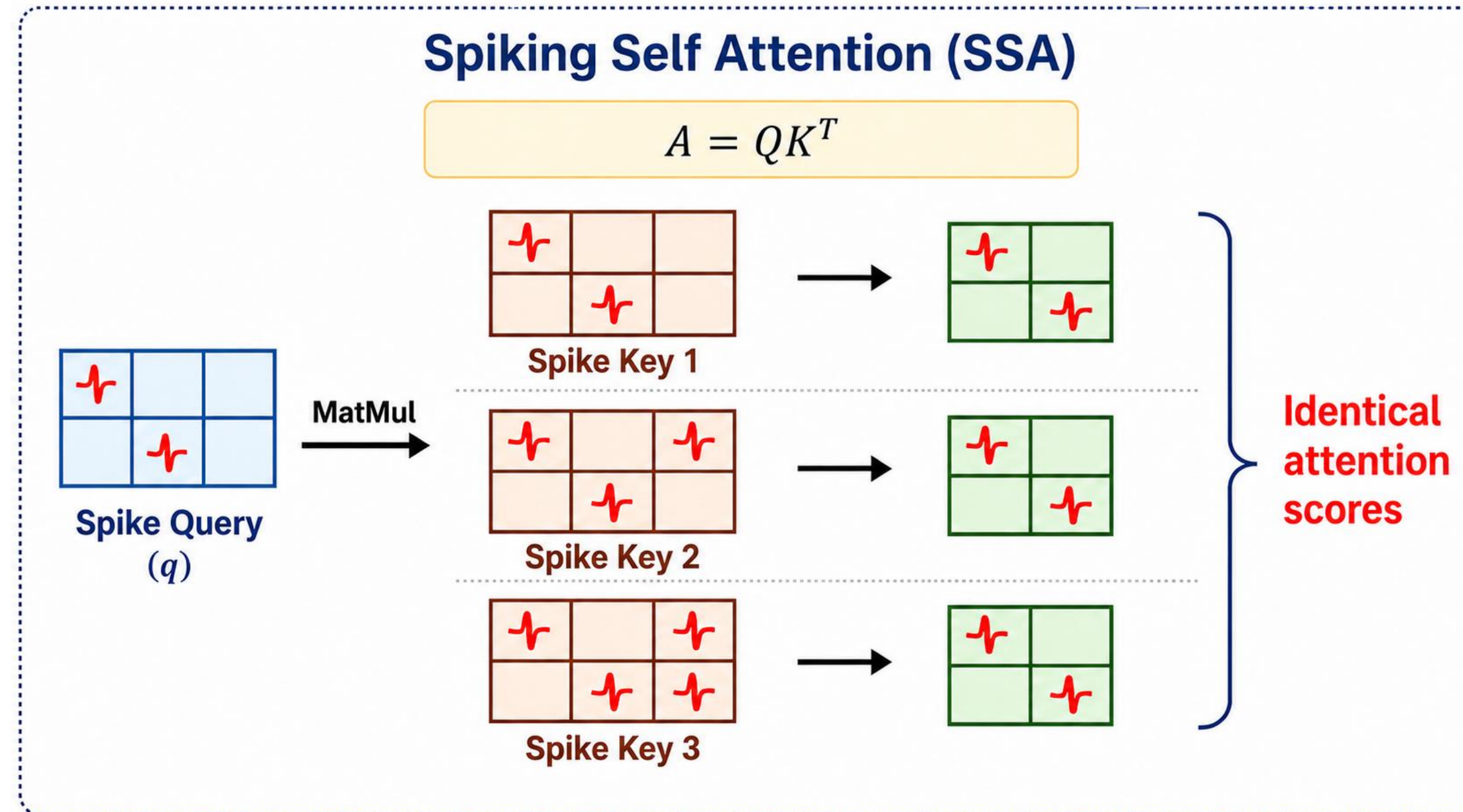


TL;DR

- We identify **density bias** in existing spiking attention.
- To address this issue, we propose **Spike Dice Attention**, which measures density-aware set similarity between binary spike patterns.
- DiceFormer, an audio-focused Spiking Transformer, achieves **SNN SOTA performance on AudioSet-20K**, delivering high accuracy and superior energy efficiency.

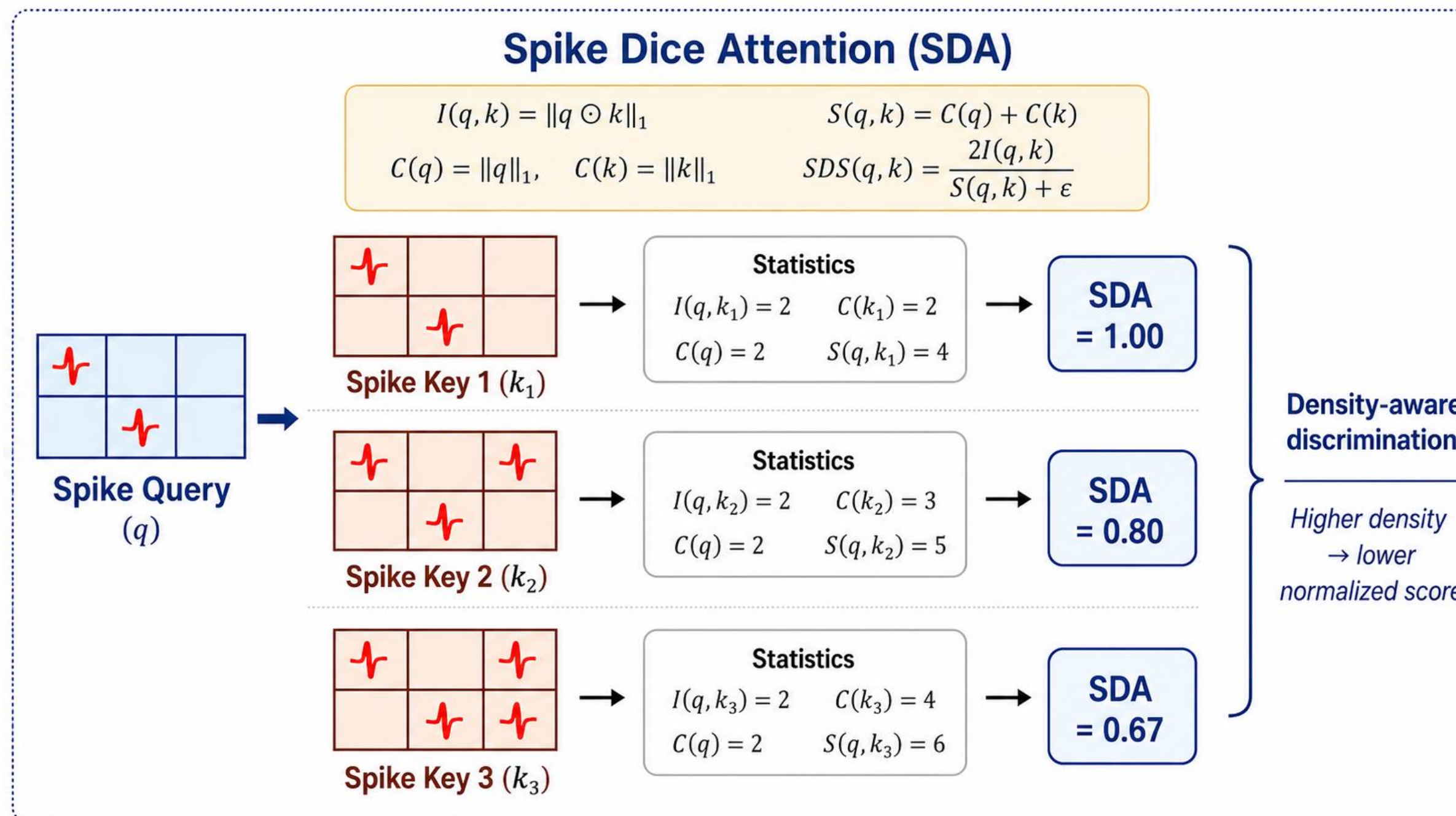
What is Density Bias?

Spiking Self Attention (SSA)



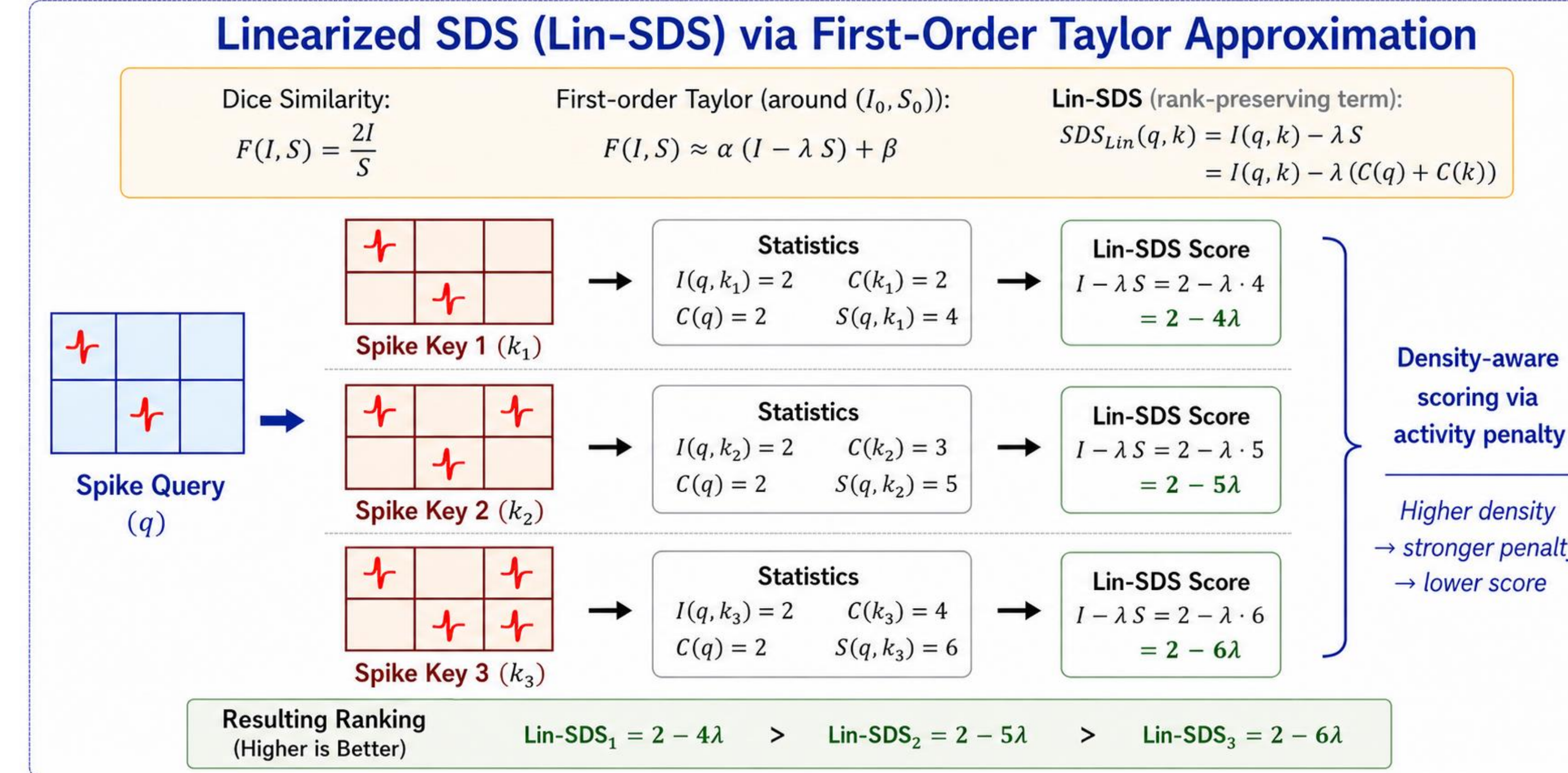
- Density bias:** For the same query, distinct keys with different spike patterns and firing densities can yield identical similarity scores.

Spike Dice Attention (SDA)



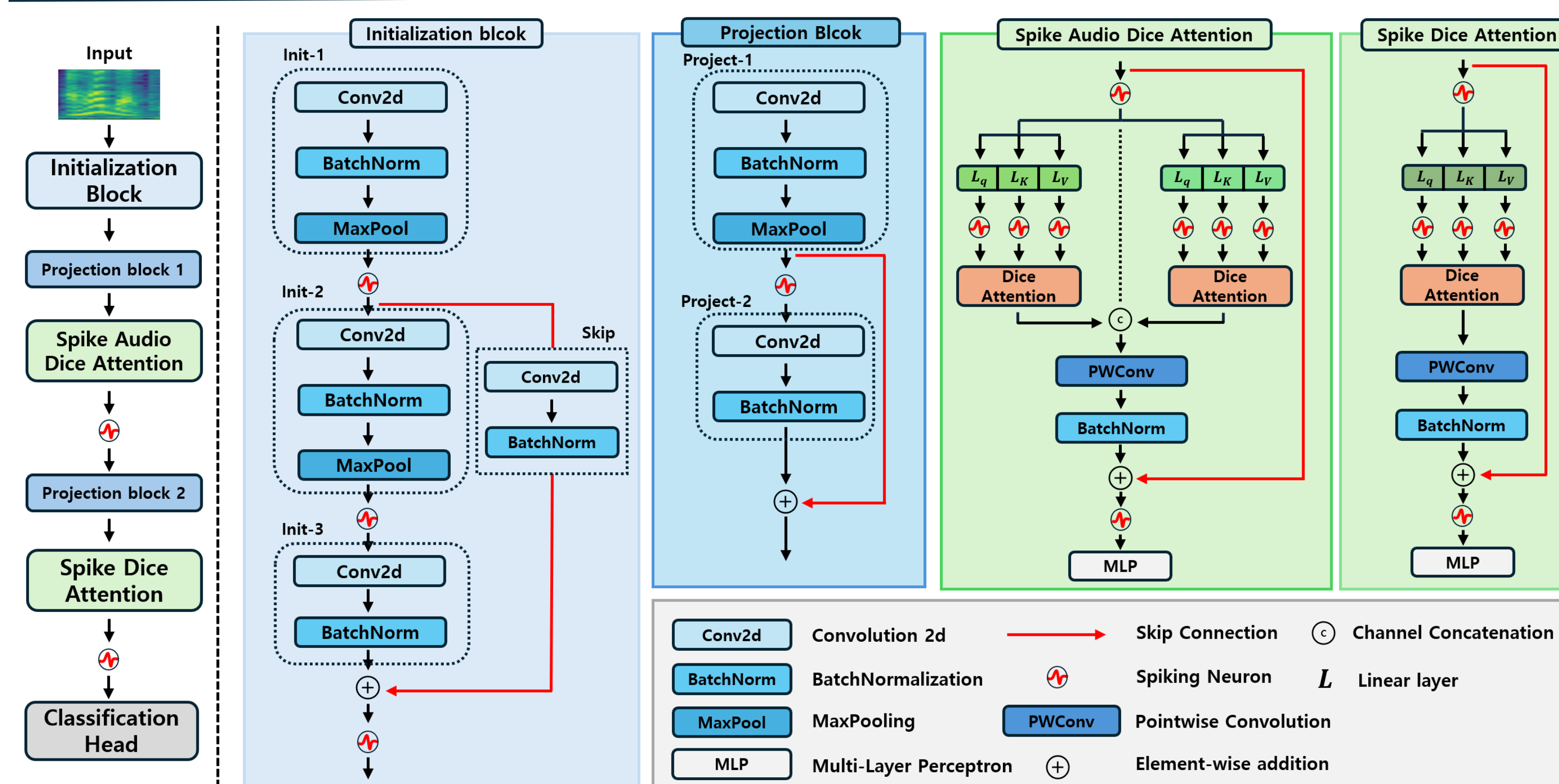
- Compute spike overlap:** $I(q, k) = |q \odot k|_1$ to measure the number of matched spike activations between query and key.
- Estimate total activity:** $S(q, k) = C(q) + C(k)$ using the spike counts of the query and key.
- Normalize by total activity:** $SDS(q, k) = \frac{2I(q, k)}{S(q, k) + \epsilon}$ yields density-aware scores that penalize dense spike patterns with unmatched activations.

Linearized Spike Dice Attention (Lin-SDA)



- Linearize Dice normalization:** Apply a first-order Taylor approximation to convert the division-based SDS into an affine form.
- Preserve score ranking:** Remove the positive scaling factor and constant offset, retaining the rank-preserving term $I(q, k) - \lambda[C(q) + C(k)]$.
- Penalize spike activity:** Use the learnable factor λ to suppress excess activity, yielding a density-aware, division-free similarity score.

Architecture



- Initialization Block:** Projects the input spectrogram from C to E channels and downsamples the spatial resolution by $4\times$.
- Projection Blocks:** Increase the feature dimension while progressively downsampling the time-frequency resolution.
- SADA Block:** Splits the features into frequency and temporal streams and applies separate Dice attention along each axis.
- SDA Block:** Flattens the downsampled feature map and performs unified Dice self-attention over all time-frequency tokens.
- Classification Head:** Aggregates the final features using global average pooling and produces the class prediction.

Results

Main Results

Model	Type	Direct training	Parameters (M)	Energy (mJ)	Time step	Score
AudioSet (20k) — mAP						
AST (Gong et al., 2021)	ANN	-	88.1	475.64	-	0.148
SSAST-S (Gong et al., 2022)	ANN	-	23	176.82	-	0.165
DTF-AT (Alex et al., 2024)	ANN	-	69	153.18	-	0.187
Spikformer (Zhou et al., 2023)	SNN	✓	65.9	18.82	4	0.136
Spike-driven Transformer (Yao et al., 2023a)	SNN	✓	65.9	8.15	4	0.130
Spike-driven Transformer V2 (Yao et al., 2024)	SNN	✓	55.0	14.92	4	0.138
QKFormer (Zhou et al., 2024)	SNN	✓	64.5	43.43	4	0.147
DiceFormer-10-S (Ours)	SNN	✓	13.7	5.34	4	0.145
DiceFormer-10-M (Ours)	SNN	✓	24.2	9.55	4	0.157
DiceFormer-10-L (Ours)	SNN	✓	54.3	6.18	1	0.153
DiceFormer-10-L (Ours)	SNN	✓	54.3	17.80	4	0.161
ESC-50 — Acc (%)						
AST (Gong et al., 2021)	ANN	-	87.2	260.54	-	88.7*
SSAST-S (Gong et al., 2022)	ANN	-	23	72.63	-	85.4*
DTF-AT (Alex et al., 2024)	ANN	-	68.6	77.28	-	89.19*
DiceFormer-10-S (Ours)	SNN	✓	13.5	5.17	4	85.37
DiceFormer-10-M (Ours)	SNN	✓	24.0	7.95	4	85.47
SCV2 — Acc (%)						
AST (Gong et al., 2021)	ANN	-	86.9	44.11	-	98.11*
SSAST-S (Gong et al., 2022)	ANN	-	23	11.32	-	97.70*
DTF-AT (Alex et al., 2024)	ANN	-	68.6	19.32	-	98.30*
DCLS-Delays (Hammouamri et al., 2024)	SNN	✓	2.5	-	-	95.35
SDC-KWS (Lim & Kim, 2025b)	SNN	✓	0.4	-	8	94.70
DiceFormer-10-S (Ours)	SNN	✓	13.5	5.31	4	97.27

Ablation Study

Methods	Time step	SADA:SDA ratio	mAP
ANALYSIS OF ATTENTION COMPONENTS			
DiceFormer-10-L (SDSA (Yao et al., 2023a) only)	4	5:5	0.126
DiceFormer-10-L (SSA (Zhou et al., 2023) only)	4	5:5	0.142
DiceFormer-10-L (SADA only)	4	5:5	0.157
DiceFormer-10-L (SDA only)	4	5:5	0.158
DiceFormer-10-L (SDA → SADA)	4	5:5	0.157
DiceFormer-10-L (SADA → SDA)	4	6:4	0.160
DiceFormer-10-L (SADA → SDA)	4	4:6	0.155
DiceFormer-10-L (SADA → SDA, Full Model)	4	5:5	0.161
EFFECTIVENESS OF BINARIZED ATTENTION MAP			
DiceFormer-10-L (SDS-continuous)	4	5:5	0.160
DiceFormer-10-L (SDS-binarized)	4	5:5	0.161
EFFECTIVENESS OF LIN-SDA			
DiceFormer-10-S (Lin-SDA)	4	5:5	0.143
DiceFormer-10-M (Lin-SDA)	4	5:5	0.154
DiceFormer-10-L (Lin-SDA)	4	5:5	0.159
GENERALIZATION OF SDA			
Spike-driven Transformer (baseline)	4	-	0.130
Spike-driven Transformer (+SDA)	4	-	0.142 (+0.012)
Spike-driven Transformer V2 (baseline)	4	-	0.138
Spike-driven Transformer V2 (+SDA)	4	-	0.144 (+0.006)
QKFormer (baseline)	4	-	0.147
QKFormer (+SDA)	4	-	0.149 (+0.002)

Conclusion

- Density-Aware Spiking Audio Transformer:** DiceFormer combines SDA and SADA to reduce spike-density bias and capture audio-specific patterns, achieving SNN SOTA on AudioSet-20K and ESC-50.
- Efficient Linearized Attention:** Lin-SDA achieves performance comparable to SDA at lower computational cost.
- Vision Applicability:** CIFAR-100 experiments confirm that SDA can be applied to spiking vision Transformer V2 backbones.



Contact